

Bluestock Mutual Fund Analytics Platform

Final Capstone Report

Prepared By	Ganesh Garad
Company	Bluestock Fintech
Domain	Mutual Fund / Fintech Analytics
Tools	Python, Pandas, SQLite, SQL, Power BI
Period	June 2026

This report presents a complete end-to-end analytics solution covering data ingestion, ETL, SQLite schema design, EDA, performance analytics, risk analytics, and dashboard reporting for mutual fund decision support.

1. Executive Summary

The project integrates 40 mutual fund schemes, 46,000 NAV observations, 32,778 investor transactions, AUM records, SIP flows, portfolio holdings, and benchmark index history into a single analytics workflow.

The latest SIP inflow tracked in the dataset is Rs 31,002 Cr and folio count reaches 26.12 Cr, highlighting strong retail participation. The latest industry AUM snapshot is approximately Rs 62.74 lakh Cr.

- Built a repeatable Python-based ETL pipeline for 10 source datasets and NAV API integration.
- Created cleaned CSV files and a SQLite database suitable for analytical queries.
- Developed EDA charts for NAV trends, AUM growth, SIP inflows, category inflows, investor demographics, and geographic distribution.
- Calculated performance metrics including CAGR, Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, tracking error, VaR, CVaR, and HHI concentration.
- Prepared dashboard assets and final presentation material for business communication.

2. Project Objectives

- Understand AMFI mutual fund data structure and public APIs.
- Prepare analysis-ready datasets from raw CSV files.
- Design a star schema for fund, date, NAV, transaction, performance, and AUM analytics.
- Create visual and statistical insights for fund comparison and investor behavior.
- Build final documentation that can be submitted as a capstone project.

3. Data Sources

Dataset	Rows	Columns	Business Use
Fund Master	40	15	Scheme metadata and classification
NAV History	46000	3	Daily NAV by AMFI code
AUM by Fund House	90	5	AUM trend by AMC
Monthly SIP Inflows	48	6	SIP inflow and SIP account trend
Scheme Performance	40	19	Return, risk, rating, and expense metrics
Investor Transactions	32778	13	Investor behavior, state, age, gender, amount
Portfolio Holdings	322	8	Sector and security weights
Benchmark Indices	8050	3	Nifty 50 and Nifty 100 time series

Source references include the project CSV datasets, AMFI-style scheme metadata, public mutual fund NAV API workflows, and benchmark index files supplied for the capstone exercise.

4. ETL Pipeline Design

- Read all raw datasets using Pandas with standardized relative paths.
- Validate shapes, dtypes, duplicate rows, missing values, and business constraints.
- Clean date fields, numeric fields, transaction enums, KYC status values, and NAV positivity.
- Export cleaned datasets into data/processed and load tables into SQLite using SQLAlchemy.
- Generate reports and notebook outputs for validation and reproducibility.

5. SQLite Star Schema

The data warehouse follows a star schema pattern where dimensions describe business entities and fact tables store measurable events.

Table	Key Fields	Purpose
dim_fund	amfi_code	Fund metadata, AMC, category, plan, risk prof
dim_date	date	Calendar date fields for trend analysis
fact_nav	amfi_code, date	Daily NAV values and daily return calculation
fact_transactions	investor_id, amfi_code, date	Investor transaction amount and type
fact_performance	amfi_code	Returns, risk ratios, Alpha/Beta, drawdown, A
fact_aum	fund_house, date	AMC-level AUM and scheme counts

6. Data Cleaning and Validation

- NAV history: parsed date, sorted by AMFI code and date, duplicate rows removed, NAV values checked to be positive.
- Investor transactions: transaction types standardized into SIP, Lumpsum, and Redemption; amount validations applied; KYC enums checked.
- Scheme performance: return columns converted to numeric values; expense ratio range checked against expected 0.1% to 2.5%.
- Processed outputs: 10 cleaned CSV files exported and SQLite row counts verified.

7. Exploratory Data Analysis - NAV and AUM

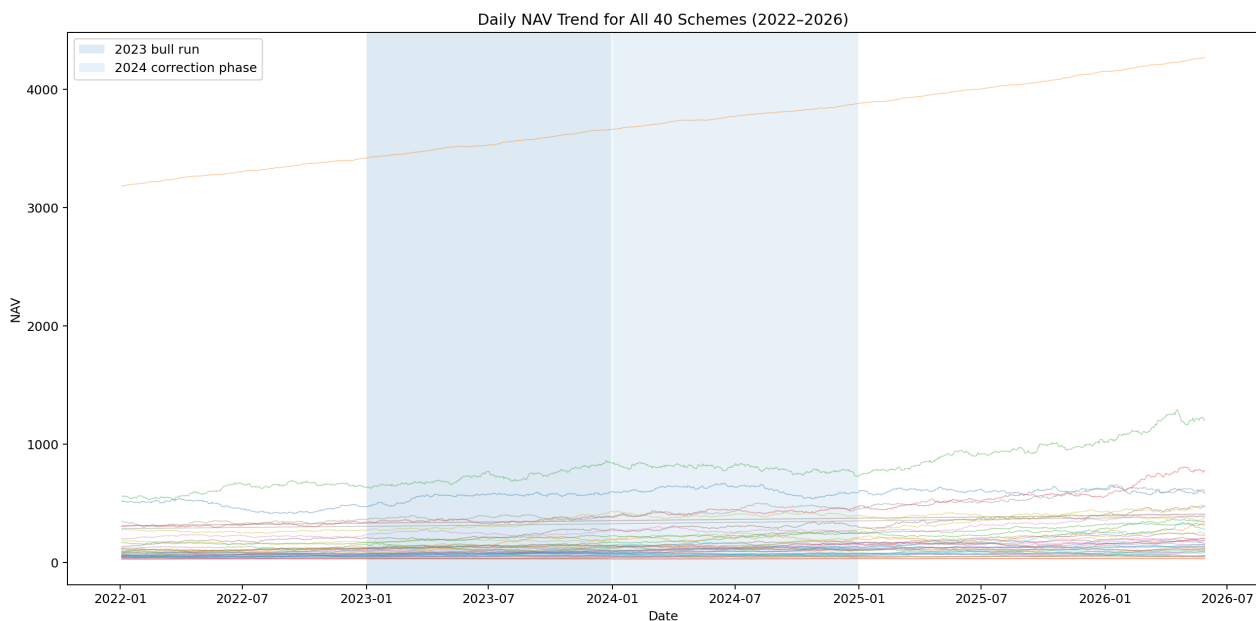


Figure 1: Daily NAV trend for all tracked schemes.

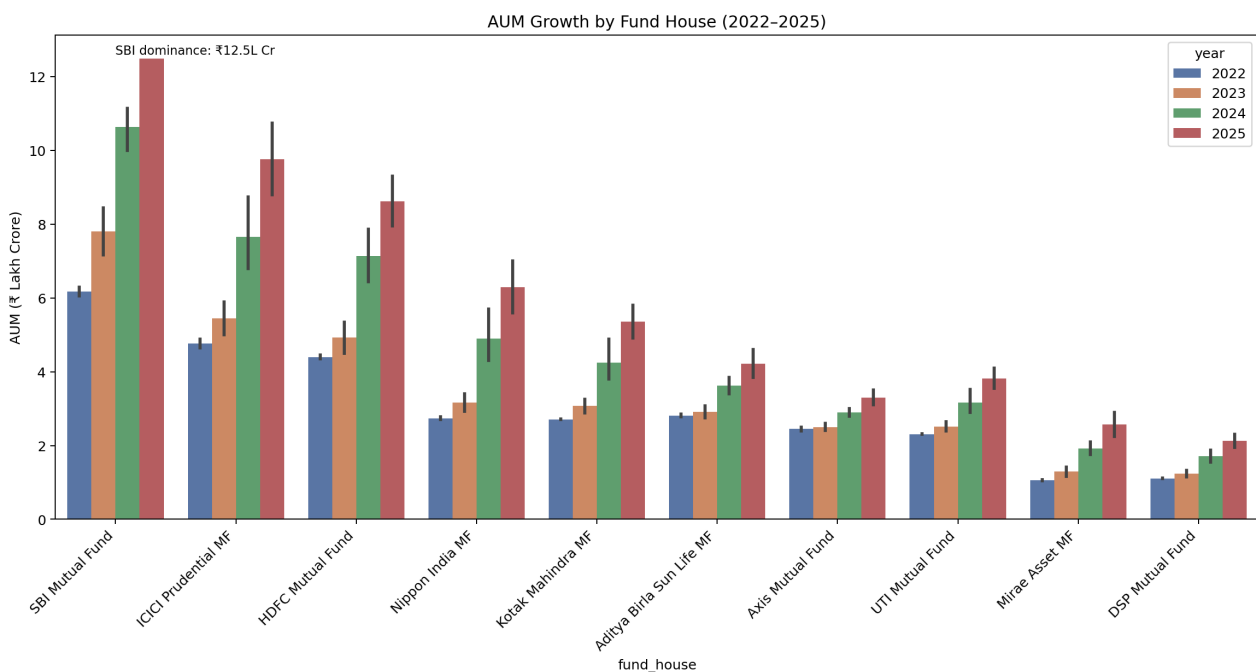


Figure 2: AUM growth by fund house. SBI Mutual Fund dominates in the latest snapshot with about Rs 12.50 lakh Cr.

8. Exploratory Data Analysis - SIP and Category Flows

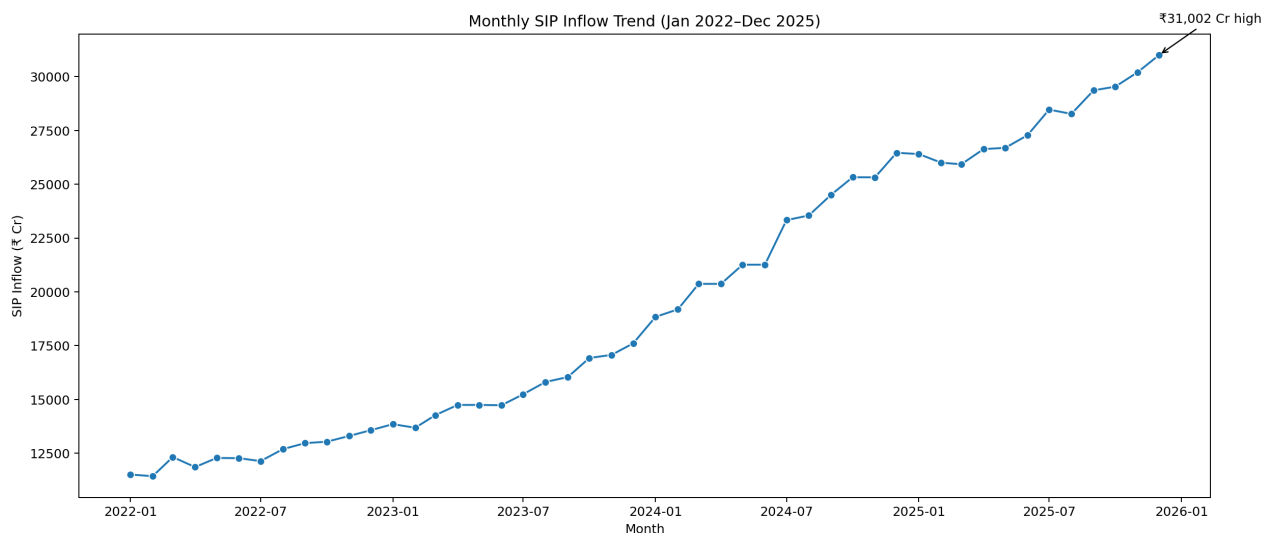


Figure 3: Monthly SIP inflow trend. Latest observed SIP inflow is Rs 31,002 Cr.

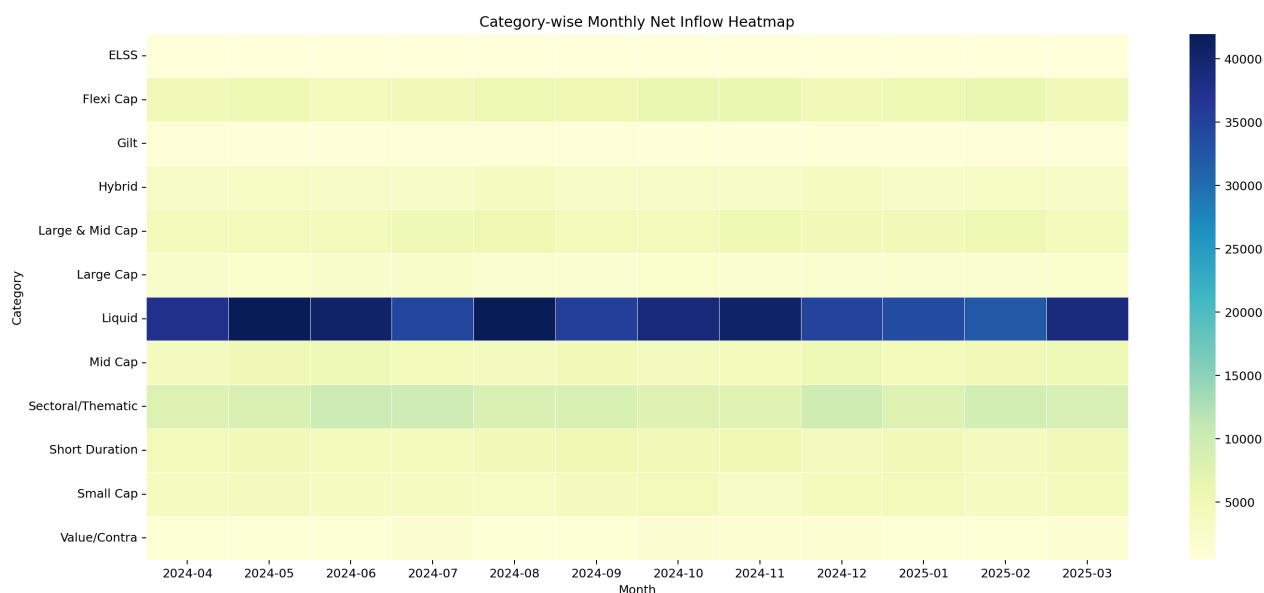


Figure 4: Category inflow heatmap by month and fund category.

9. Investor Analytics

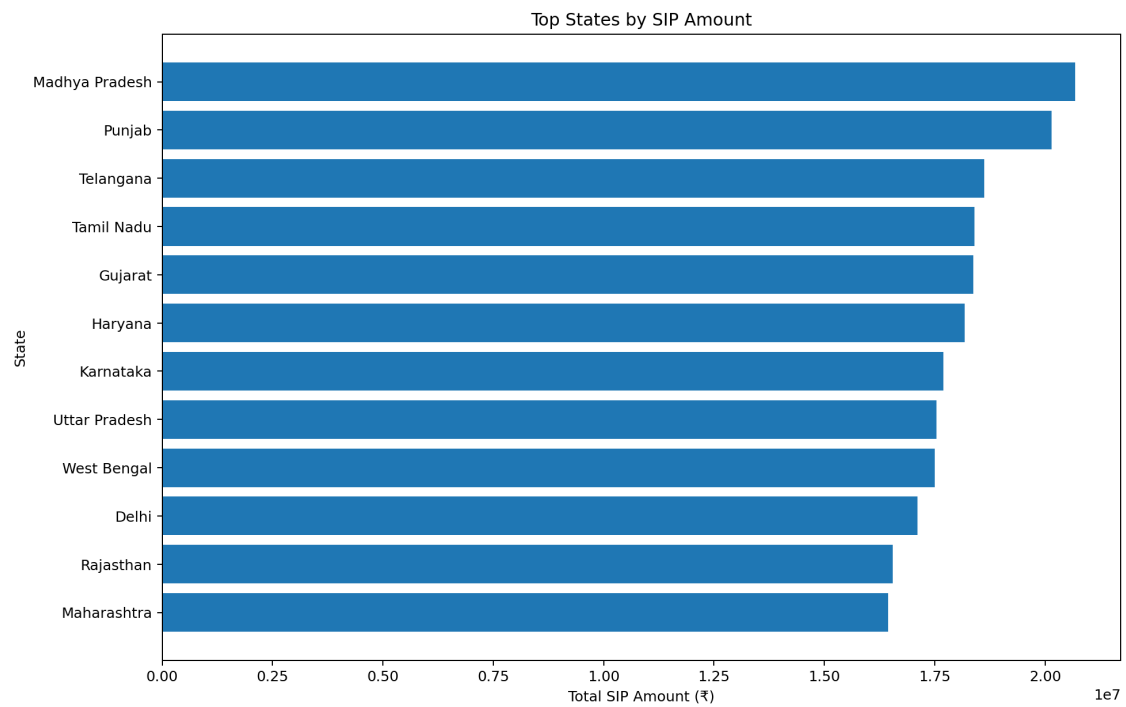


Figure 5: SIP amount by state reveals geographic concentration of investments.

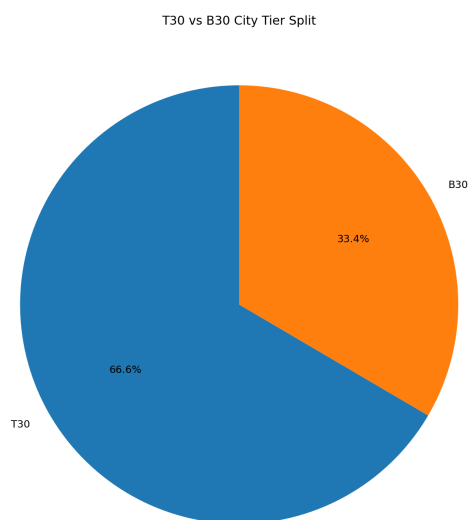


Figure 6: T30 vs B30 split supports city-tier segmentation for investor analytics.

10. Folio Growth and Correlation Analysis

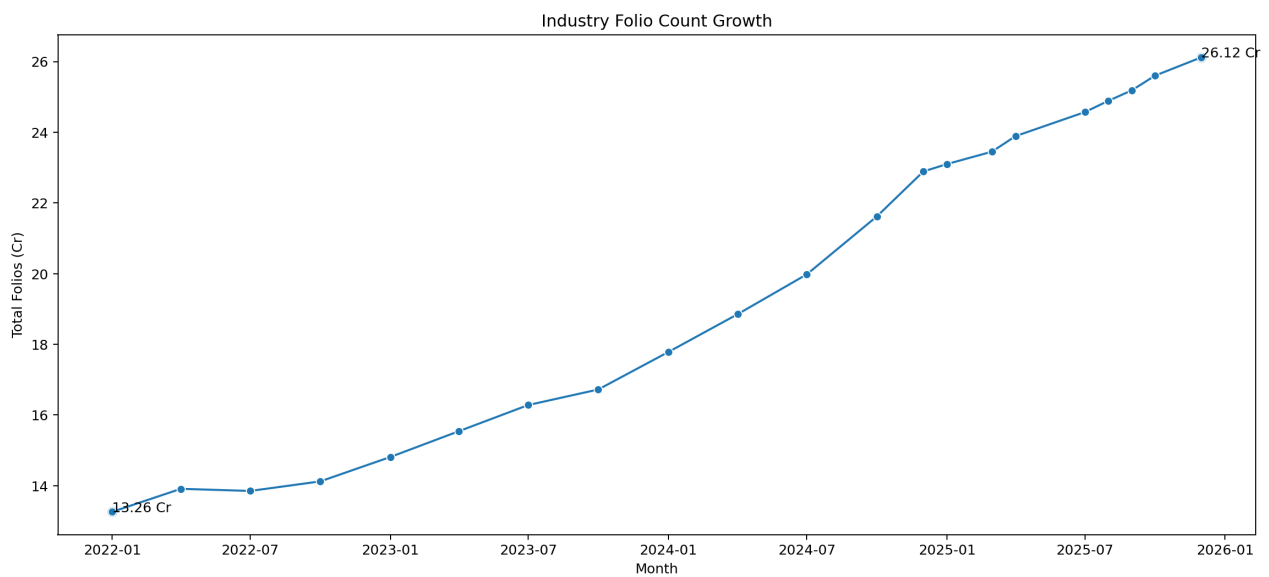


Figure 7: Folio count growth from 2022 onward reaching 26.12 Cr in the latest period.

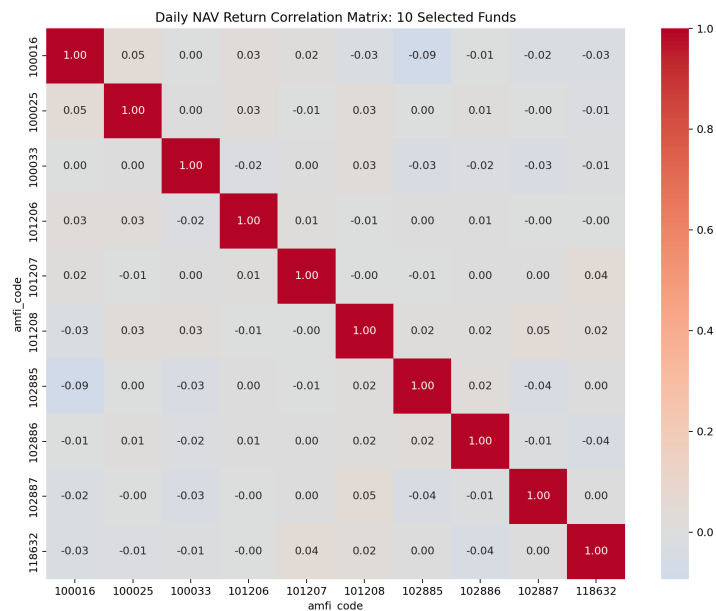


Figure 8: NAV return correlation matrix for selected funds.

11. Performance Analytics

Performance analytics compares funds on absolute return, downside risk, benchmark sensitivity, and risk-adjusted returns. The fund scorecard combines 3-year return rank, Sharpe ratio rank, Alpha rank, expense ratio rank, and drawdown rank.

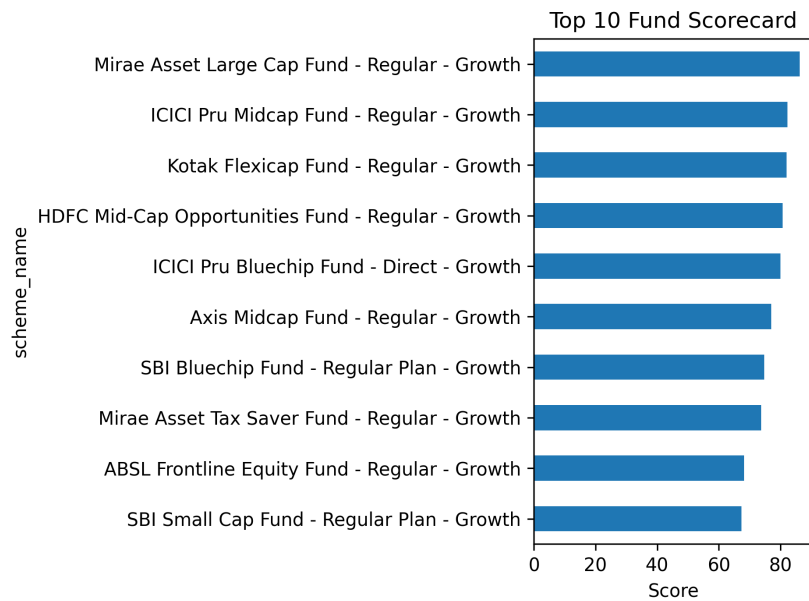


Figure 9: Top 10 funds by composite scorecard.

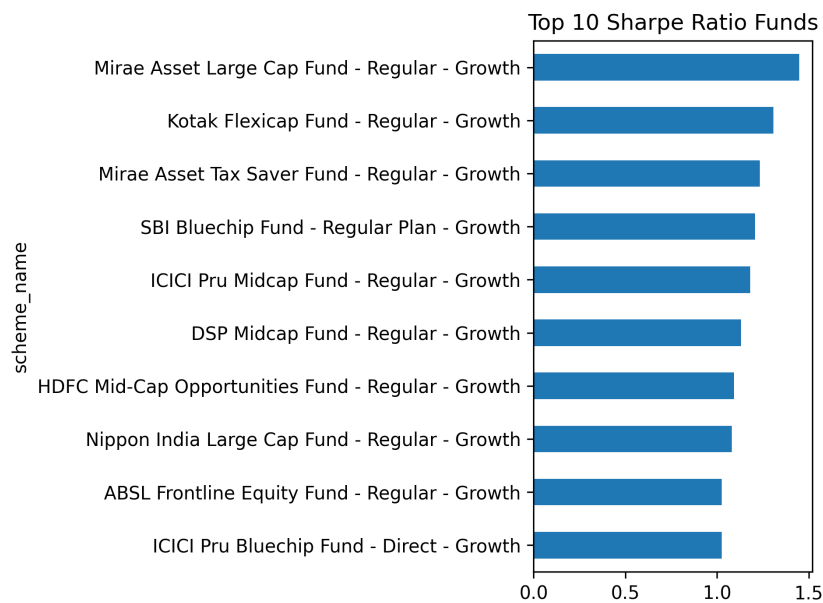


Figure 10: Top funds by Sharpe ratio.

12. Benchmark and Risk Analysis

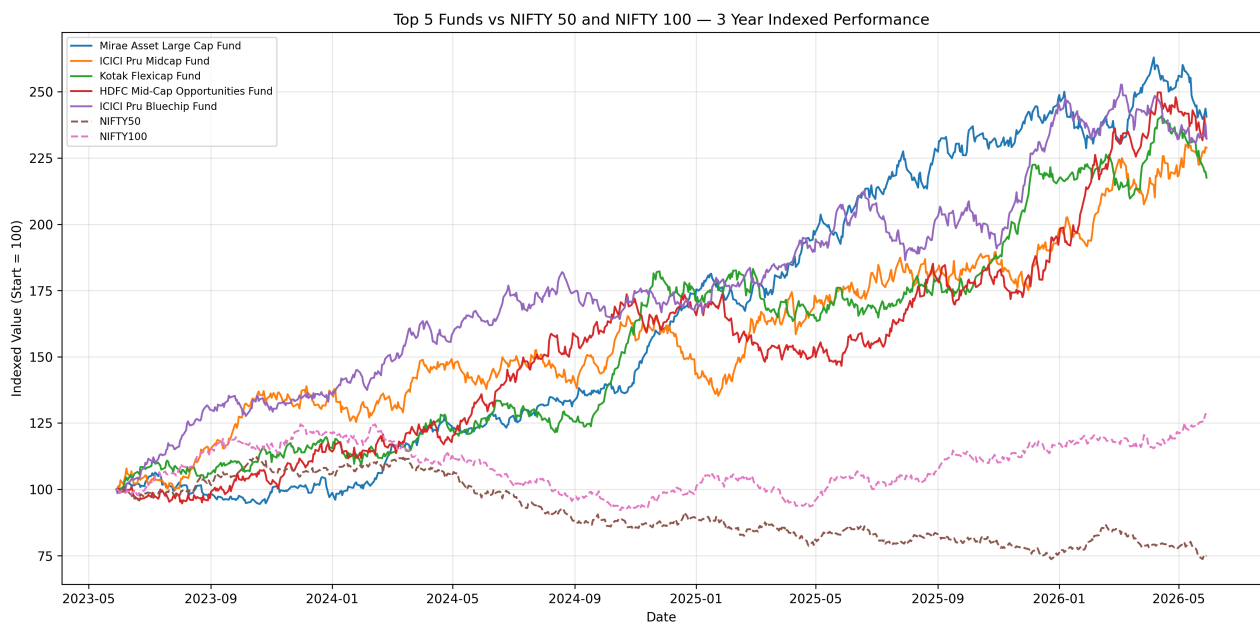


Figure 11: Top 5 funds compared with benchmark indices over three years.

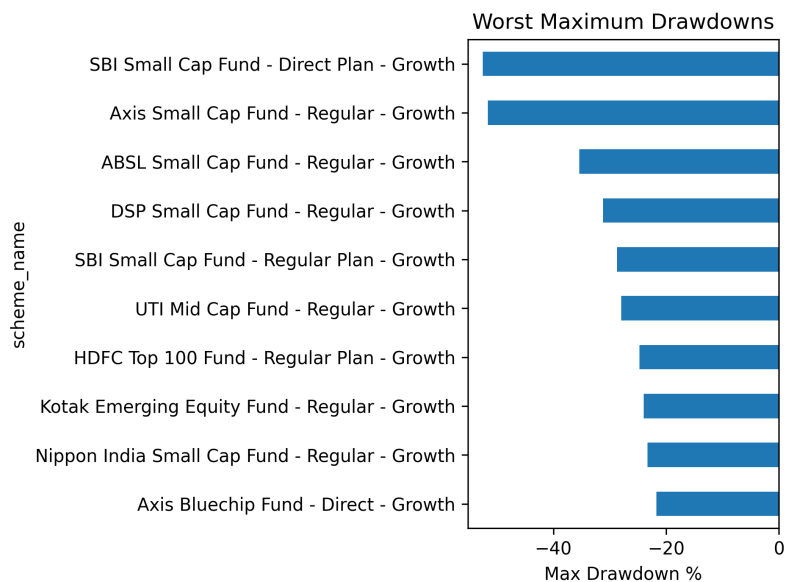


Figure 12: Maximum drawdown comparison for risk assessment.

13. Advanced Analytics

Advanced analytics adds risk distribution, downside loss behavior, investor cohort analysis, SIP continuity, recommender logic, and portfolio concentration metrics.

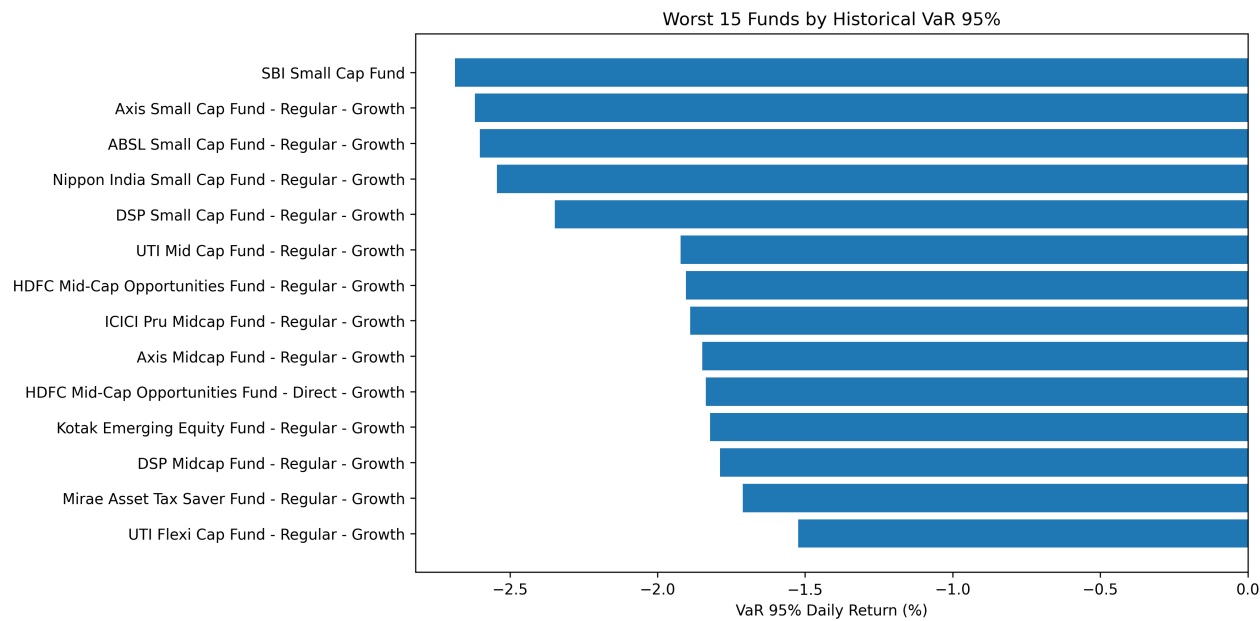


Figure 13: Worst funds by 95% historical VaR/CVaR based on daily return distribution.

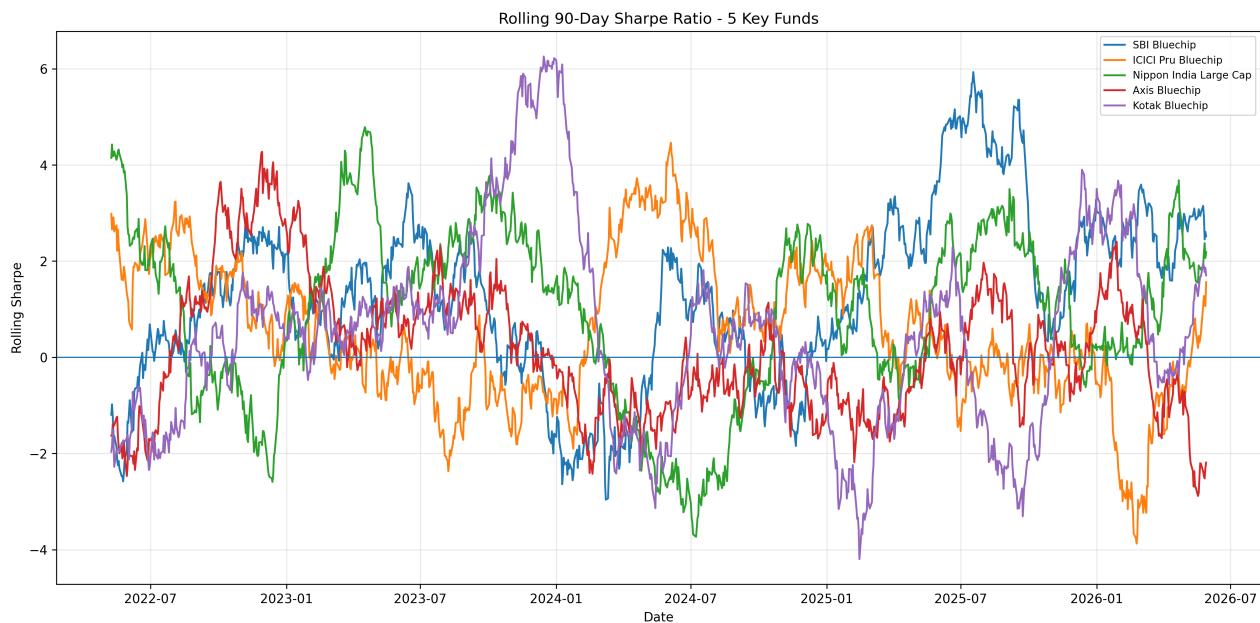


Figure 14: Rolling 90-day Sharpe ratio for key funds.

14. Investor Cohorts, SIP Continuity and HHI

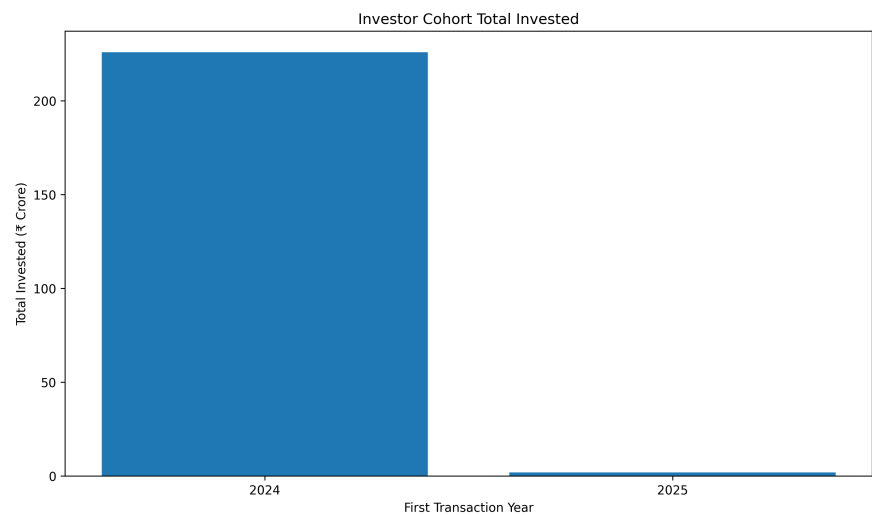


Figure 15: Investor cohort total invested by first transaction year.

SIP Continuity: Regular vs At-Risk Investors

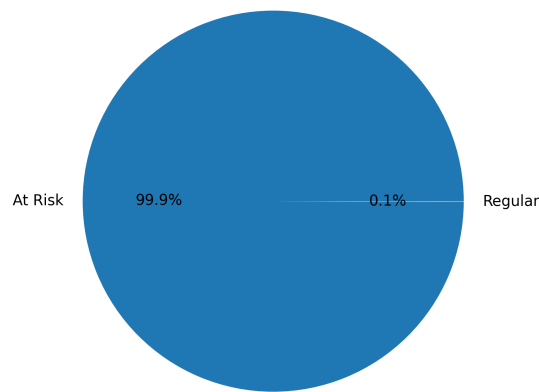


Figure 16: SIP continuity analysis classifies investors as regular or at-risk.

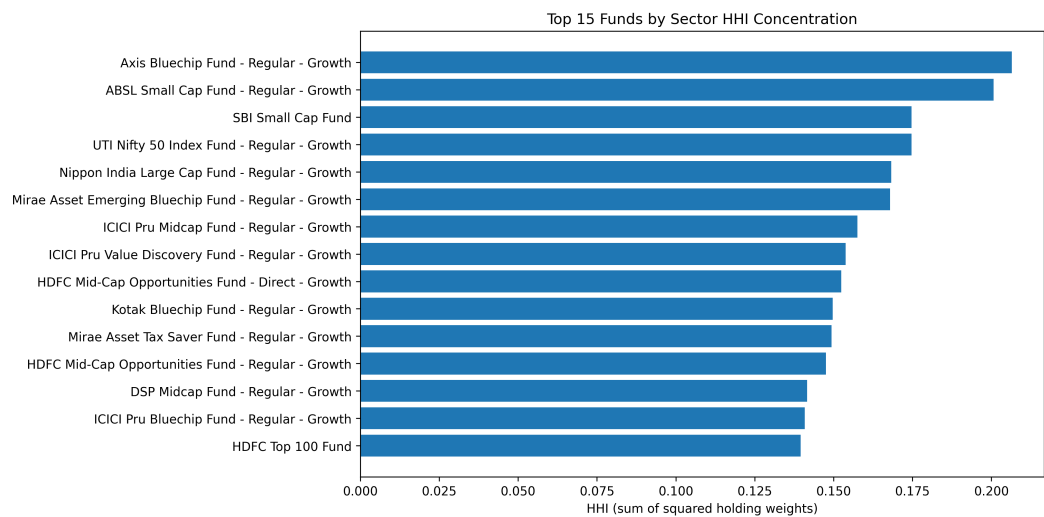
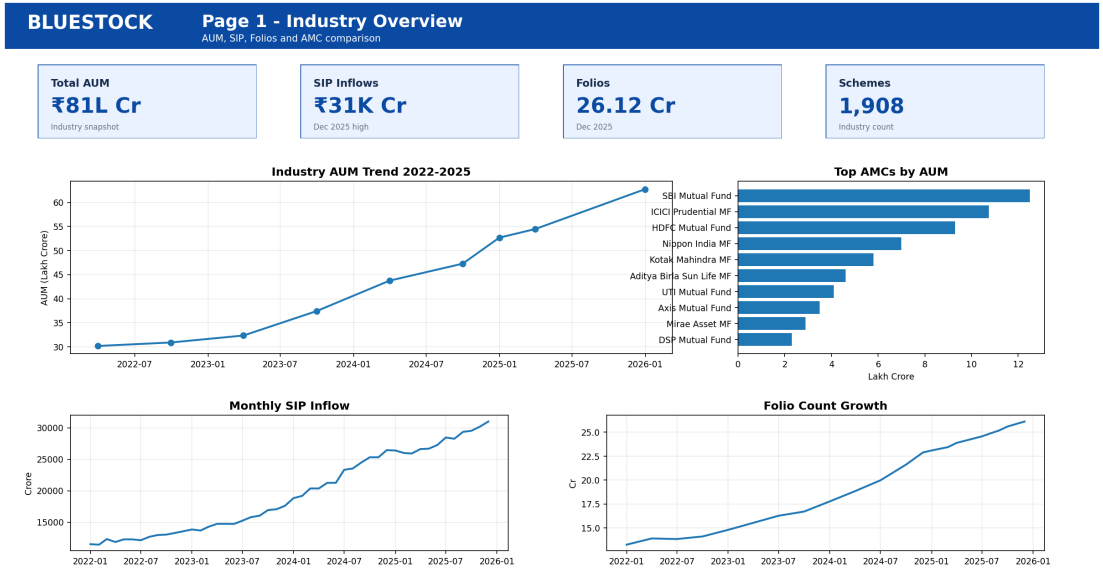


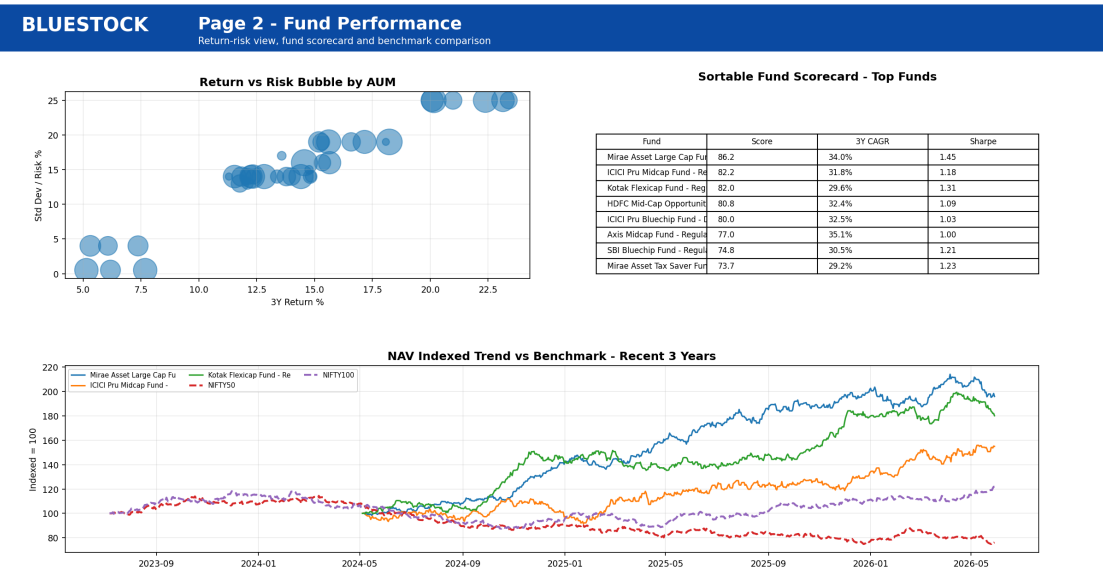
Figure 17: Sector HHI highlights concentrated equity portfolios.

15. Power BI Dashboard Overview

The dashboard is designed across four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. It uses KPI cards, slicers, fund tables, trend lines, geography views, donut charts, and benchmark comparisons.

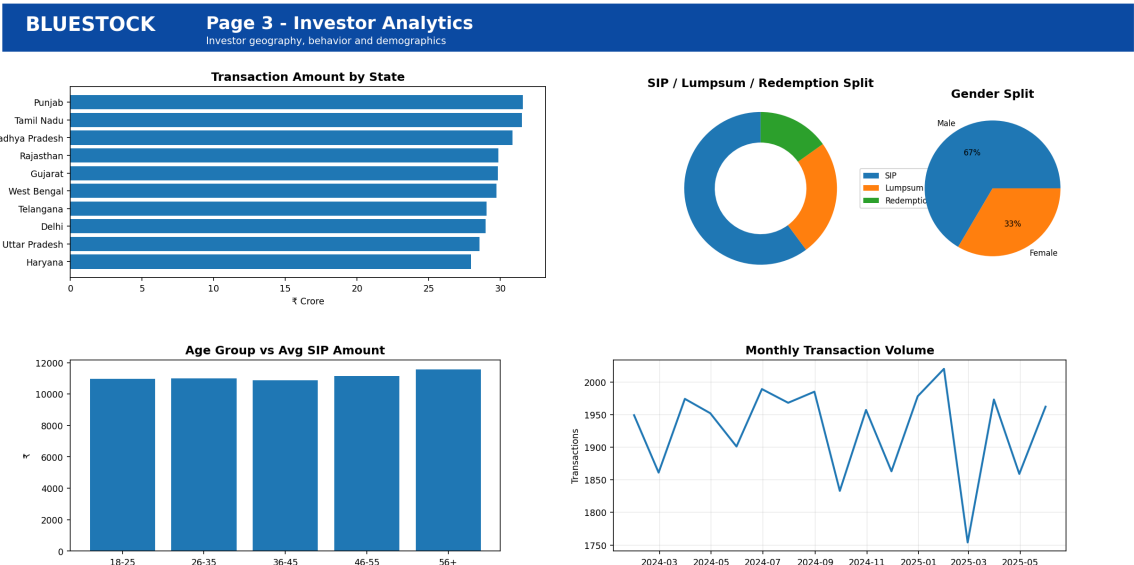


Dashboard Page 1: Industry Overview

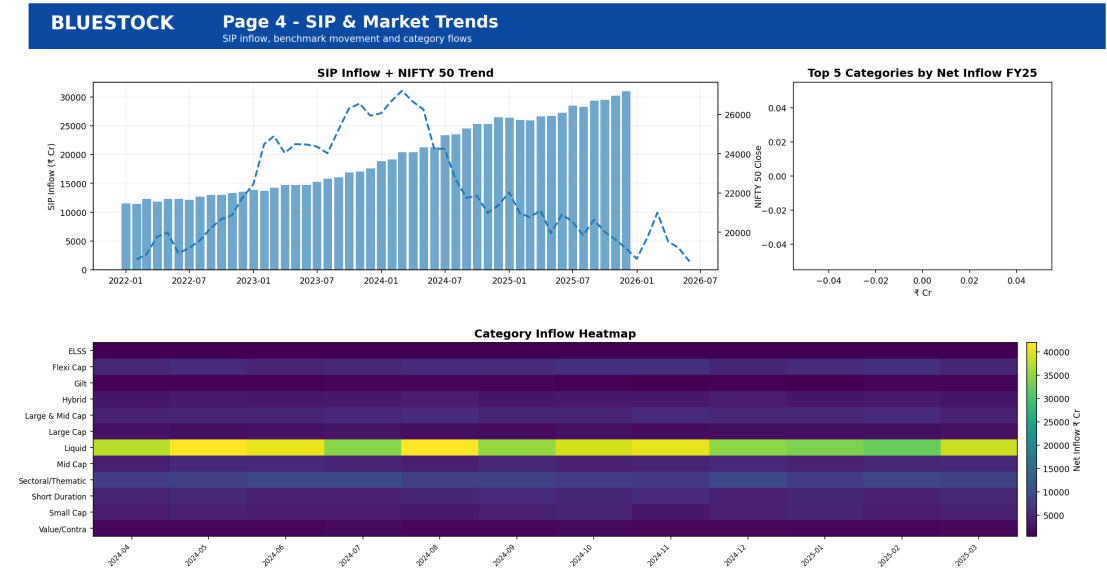


Dashboard Page 2: Fund Performance

16. Dashboard Investor and Market Pages



Dashboard Page 3: Investor Analytics



Dashboard Page 4: SIP & Market Trends

17. Key Findings

- SIP activity is strong, with latest tracked SIP inflow of Rs 31,002 Cr.
- Folio count expansion to 26.12 Cr indicates broad retail participation.
- SBI Mutual Fund leads the latest AUM snapshot among tracked fund houses.
- Direct plans generally show lower expense ratio than regular plans and are important for investor cost analysis.
- Correlation analysis shows that many equity schemes move together, so diversification needs sector and category-level review.
- VaR and CVaR identify downside risk beyond average return metrics.
- SIP continuity flags help identify investors who may need engagement or retention communication.
- Sector HHI helps identify concentrated fund portfolios where sector risk may be higher.

18. Limitations

- The dashboard package includes screenshots and build assets; the actual PBIX must be saved from Power BI Desktop.
- Source data is capstone/project data and should not be treated as live investment advice.
- Some NAV API outputs may vary over time if fetched again.
- Returns and risk measures depend on the available date range and benchmark mapping.

19. Recommendations

- Use risk-adjusted metrics together with CAGR instead of relying on return alone.
- Track SIP continuity monthly to identify at-risk investors early.
- Use category and sector exposure controls before recommending funds.
- Deploy the data model to Power BI Service or Tableau Public for shareable dashboard access.
- Add automated monthly refresh for NAV, SIP, AUM, and benchmark datasets.

20. Conclusion

The project delivers a full analytics platform from raw mutual fund datasets to final business reporting. It combines ETL, SQL, EDA, performance analytics, advanced analytics, and dashboard storytelling. The final repository is ready for submission with notebooks, scripts, reports, visualizations, SQLite database, and presentation assets.

Final Deliverables Included

- Final_Report.pdf
- Bluestock_MF_Presentation.pptx
- README.md
- run_pipeline.py
- Advanced_Analytics.ipynb
- Performance_Analytics.ipynb
- EDA_Analysis.ipynb
- SQL schema and queries
- Cleaned CSV files and SQLite database
- Dashboard PDF and screenshots

Prepared by Ganesh Garad for Bluestock Fintech Mutual Fund Analytics Capstone.